Haptics and Kinesthetic devices

3/31/2020

today's objectives

introduce you to the field of haptics (definition, why it is important, and why haptic technology is challenging)

give an overview of course content and policies

what is haptics?

haptic box



Pass it around. Feel inside.
Try to identify three objects.

hearing

sight

smell

taste

touch

which sense is most valuable to you?

which would you relinquish last?

hap·tic ('hap-tik)

adj. Of or relating to the sense of touch.

[Greek *haptikos*, from *haptesthai*, to grasp, touch. (1890)]

Cutaneous

Temperature

Texture

Slip

Vibration

Force



Johansson and Westling

Kinesthesia

Location/configuration

Motion

Force

Compliance

The haptic senses work together with the motor control system to:

- Coordinate movement
- Enable perception

what would life be like without touch?







Cutaneous

https://www.youtube.com/watch?v=0LfJ3M3Kn80



Kinesthesia

http://www.youtube.com/watch?v=FKxyJfE831Q

why do we have brains?







sea
squirt

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how does your computer/ smartphone/iPad see you?

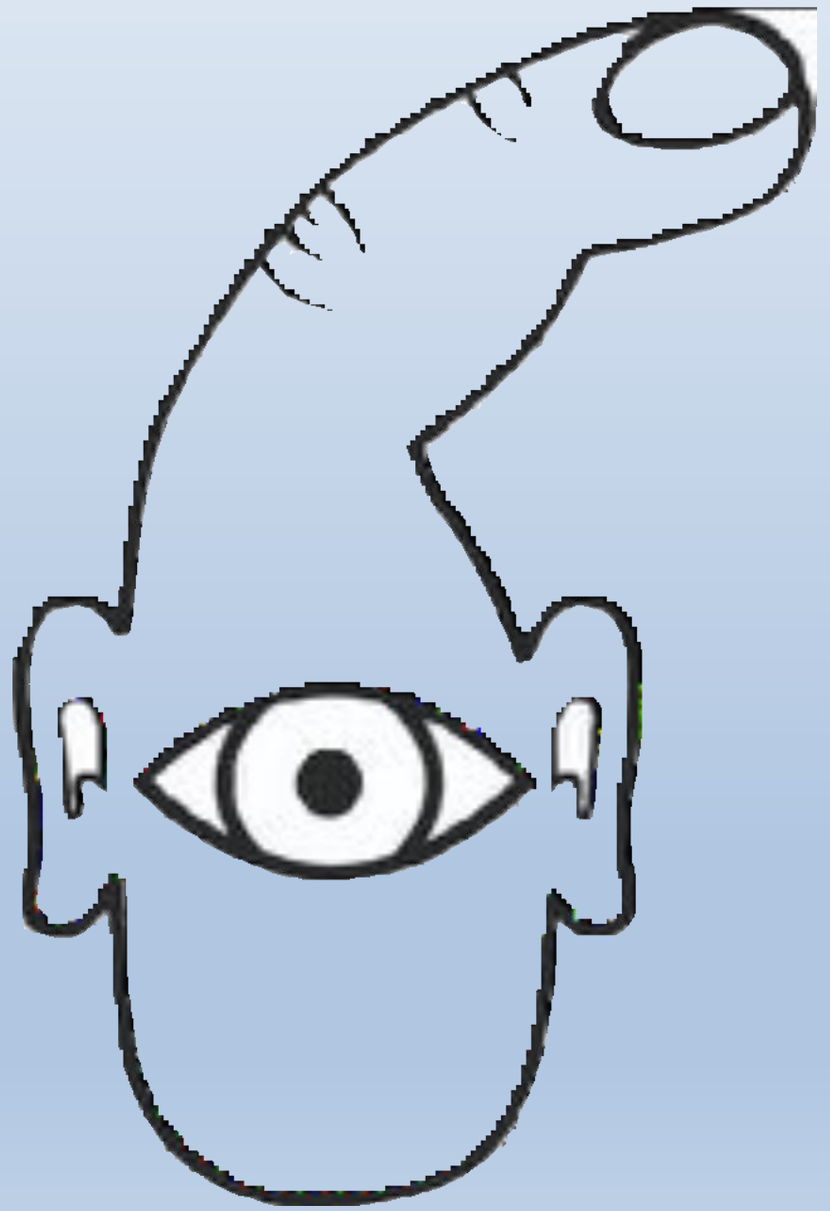


illustration by Tom Igoe

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what if...

you could make any surface
feel fuzzy, sticky, soft...?

you could touch
priceless works of art?

there was a haptic holodeck?

haptics could *teach* you?

tactual stereognosis

- Tactual = tactile = via the sense of touch
- Stereognosis = the mental perception of three-dimensionality by the senses, usually in reference to perceiving the form of solid objects by touch
- One study (Klatzky et al., 1985) showed that subjects could identify 100 common objects almost perfectly, taking about 2 seconds per object.
- People are *very good* at tactual stereognosis.

Tactile Devices



Stimulate skin to create contact sensations



Hybrid Devices

Attempt to combine tactile and kinesthetic feedback



Kinesthetic Devices

Apply forces to guide or inhibit body movement

existing applications of haptics

entertainment



Nintendo

education



Stanford/JHU



Immersion

human-computer interfaces

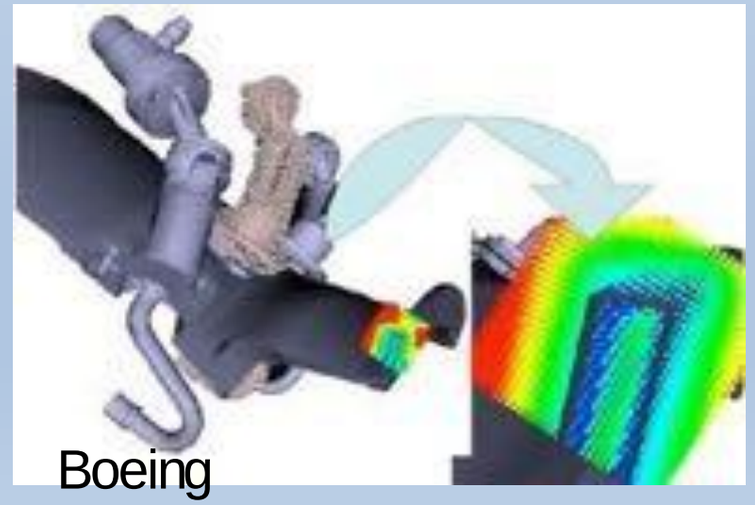




BMW



Samsung



Boeing

don’t be shy

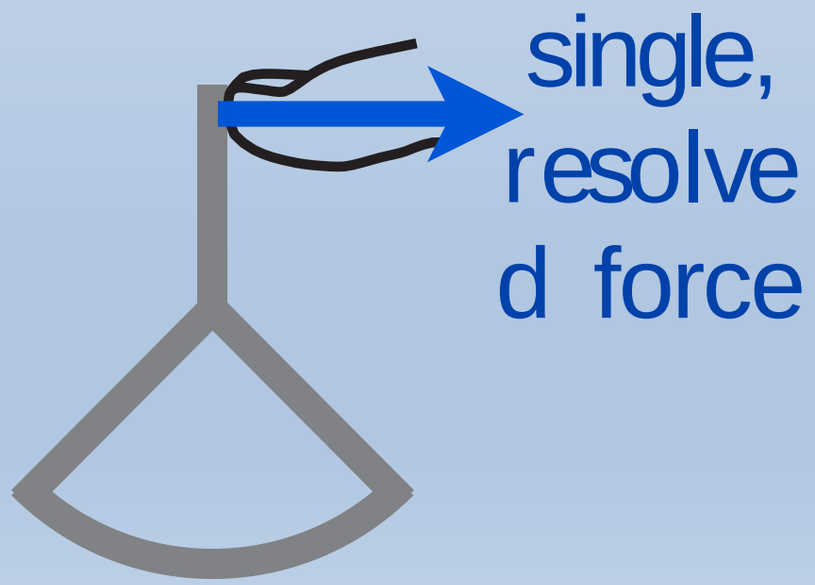
Learning Method	Retention
What one reads	10%
What one hears	26%
What one sees	30%
What one sees and hears	50%
What one speaks	70%

J.E. Stice, Engineering Education, pp. 291-296, 1987

kinesthetic (force-feedback) device basics

kinesthetic vs. tactile haptic devices

Kinesthetic haptic devices display forces or motions through a tool

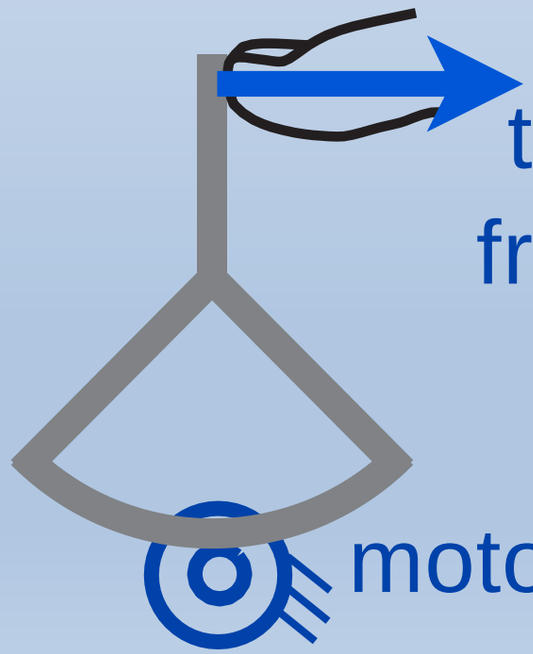


Tactile haptic devices stimulate the skin

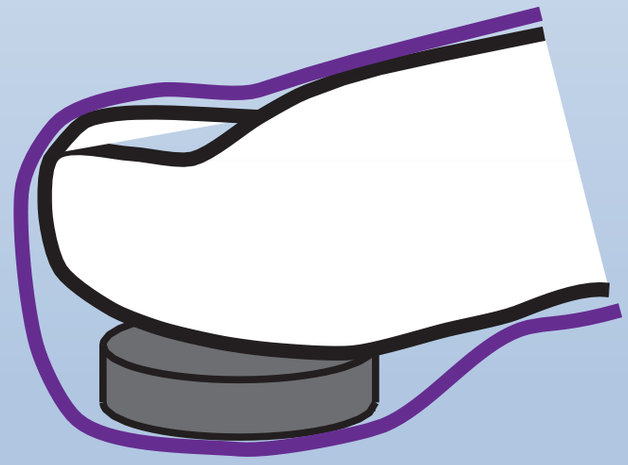


kinesthetic vs. tactile haptic devices

Kinesthetic haptic devices are usually **grounded**



Tactile haptic devices can more easily be **wearable**



vibration feedback element encased in glove

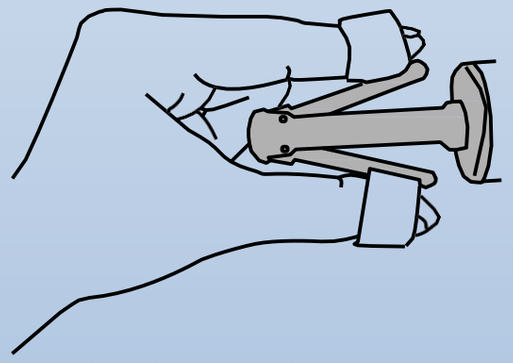
typical kinesthetic device configurations

manipulandum



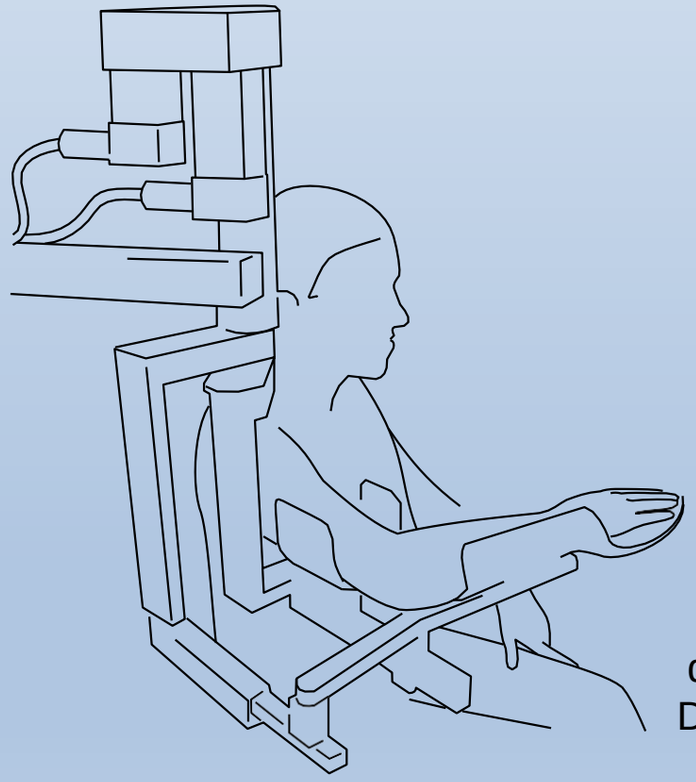
drawing by Jorge Cham

grasp



drawing by Tricia Gibo

exoskeleton



drawing by David Grow

manipulandums (expensive)



Omega
from Force Dimension

delta configuration
3 degrees of freedom



Phantom Premium 1.5
from SensAble/Geomagic

5-bar + rotation
3 degrees of freedom



Virtuose
from Haption

additional “wrist”
6 degrees of freedom

all images from Wikimedia Commons

manipulandums (cheaper)



Falcon
from Novint

delta configuration
3 degrees of freedom

image from Wikimedia Commons



Phantom Omni/Touch
from SensAble/Geomagic

5-bar + rotation
3 degrees of freedom

photographed by Akiko Nabeshima



Sidewinder
from Microsoft

spherical mechanism
2 degrees of freedom

image from Wikimedia Commons

Grip/grasp



Custom haptic gripper for Phantom Premium

© 2007 IEEE. Reprinted, with permission, from L. N. Verner and A. M. Okamura.. Effects of Translational and Gripping Force Feedback are Decoupled in a 4-Degree-of-Freedom Telemanipulator, World Haptics Conference,, pp. 286-291, 2007



Single-finger Cybergrasp from Cyberglove Systems

photograph courtesy
Stanford Center for Design Research



da Vinci Surgical System from Intuitive Surgical, Inc. (no programmable force feedback on gripper)

photographed by Akiko Nabeshima

Exoskeletons



KINARM Exoskeleton
from BKIN Technologies



Harvard



DARPA

images from Wikimedia Commons

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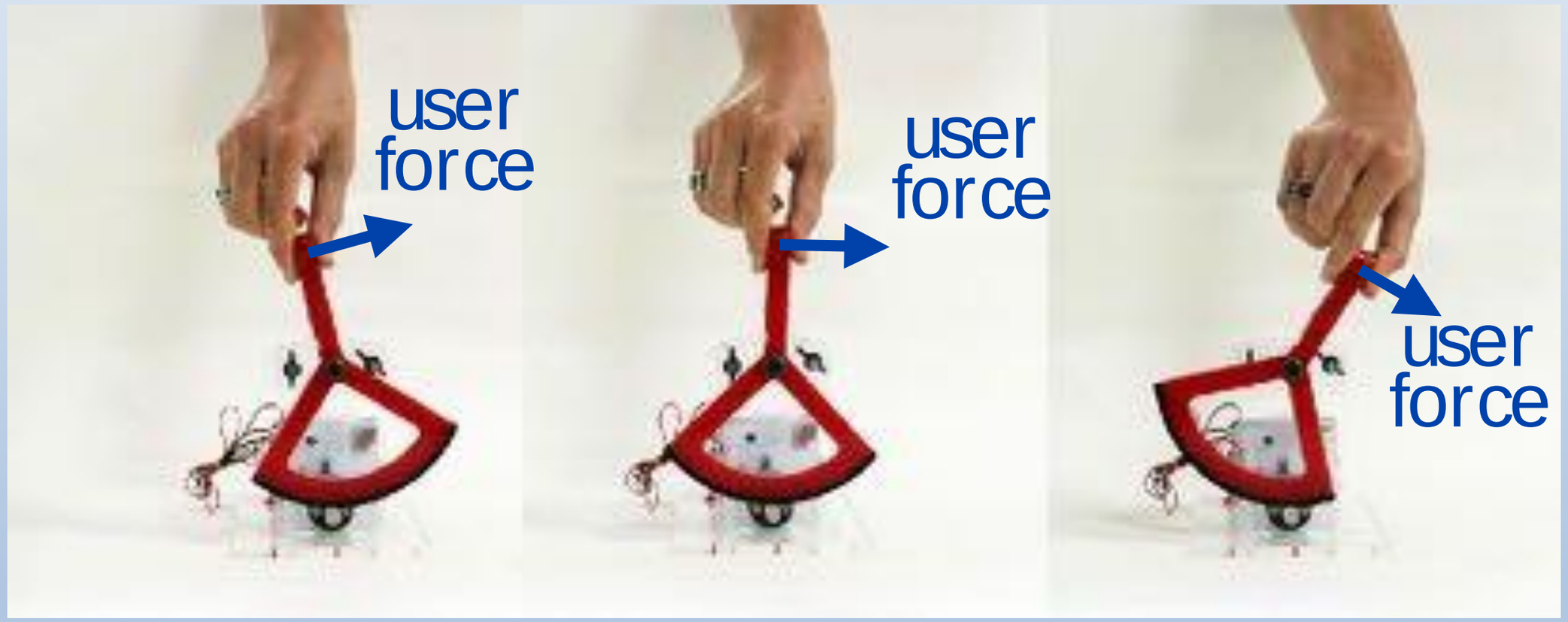
Hapkit







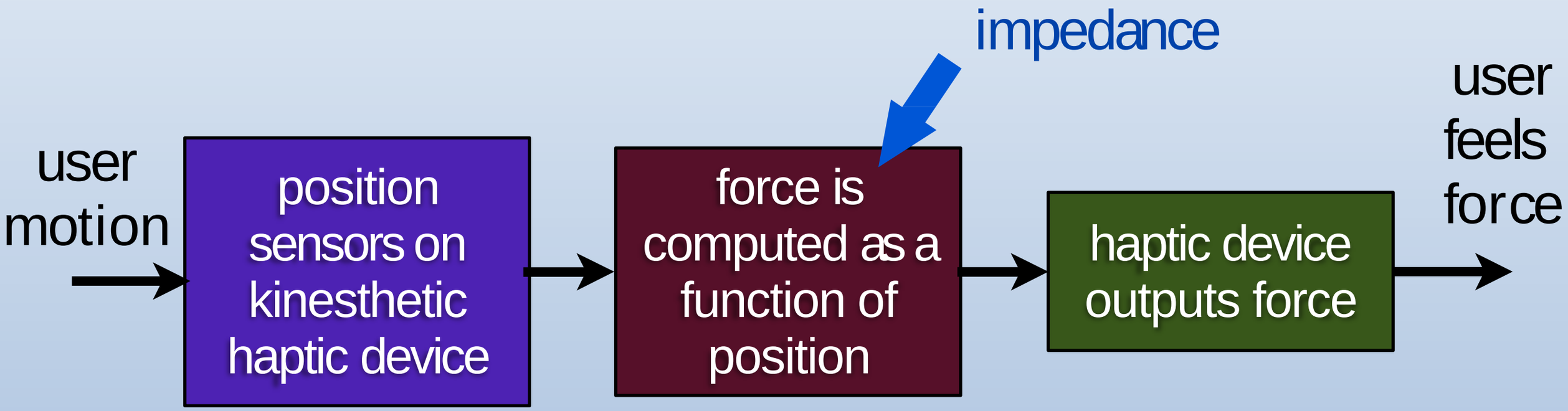
Hapkit



Hapkit

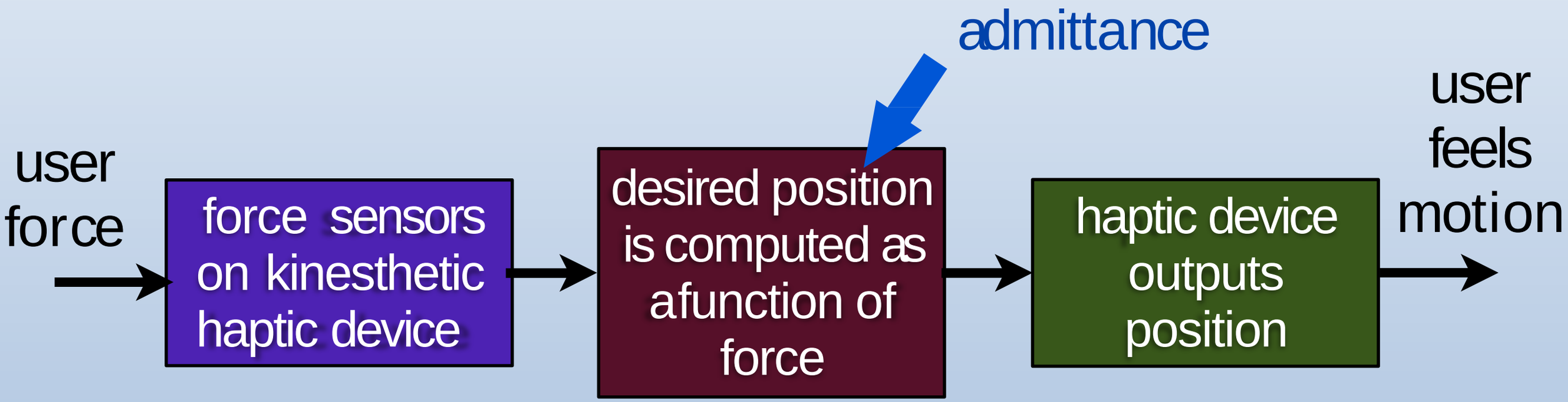


impedance-type kinesthetic devices



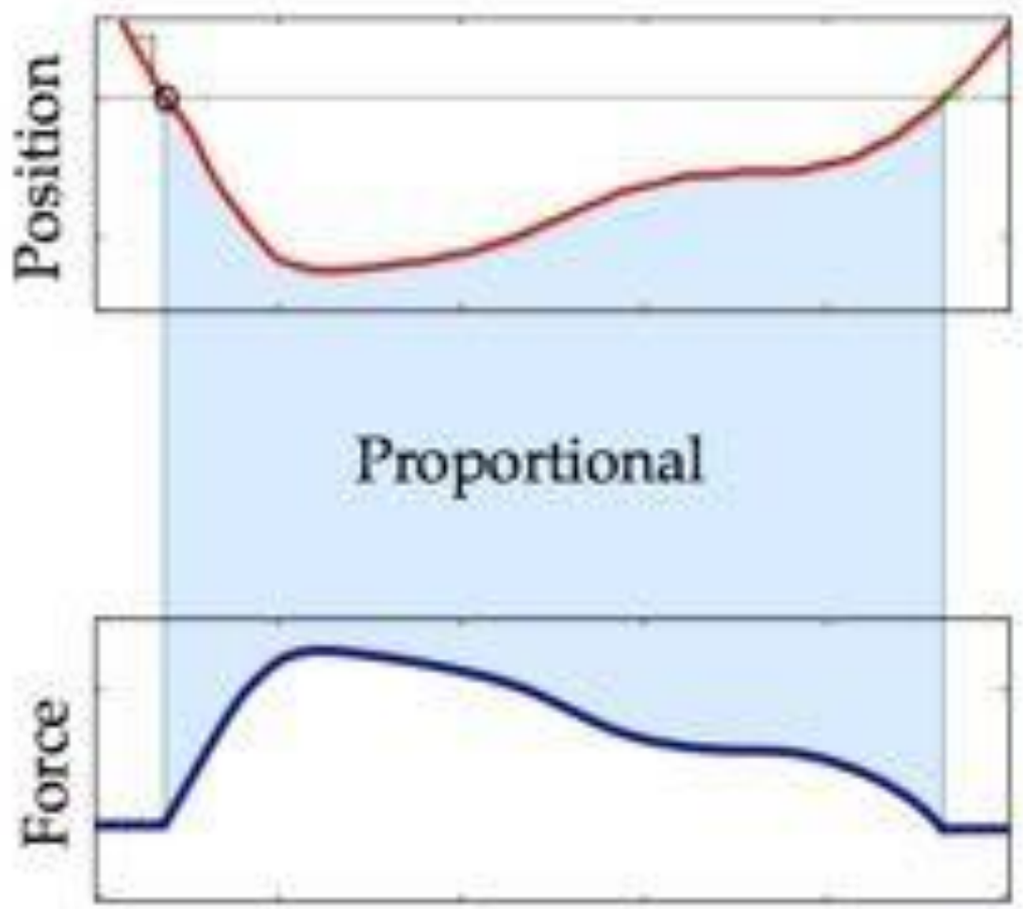
most force feedback devices are of the "impedance" type

admittance-type kinesthetic devices



“admittance”-type devices are not as common

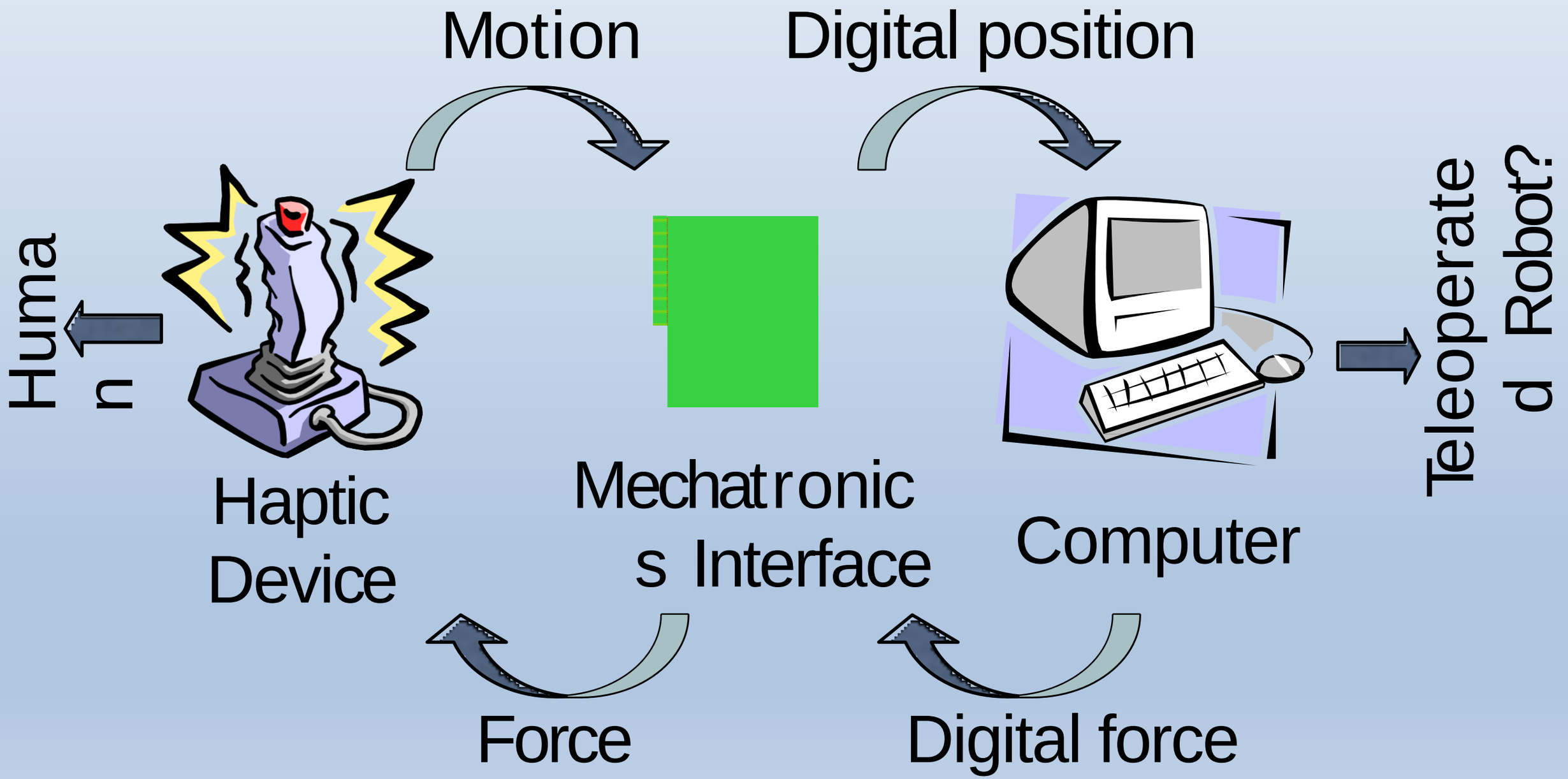



Position
Proportional
Force

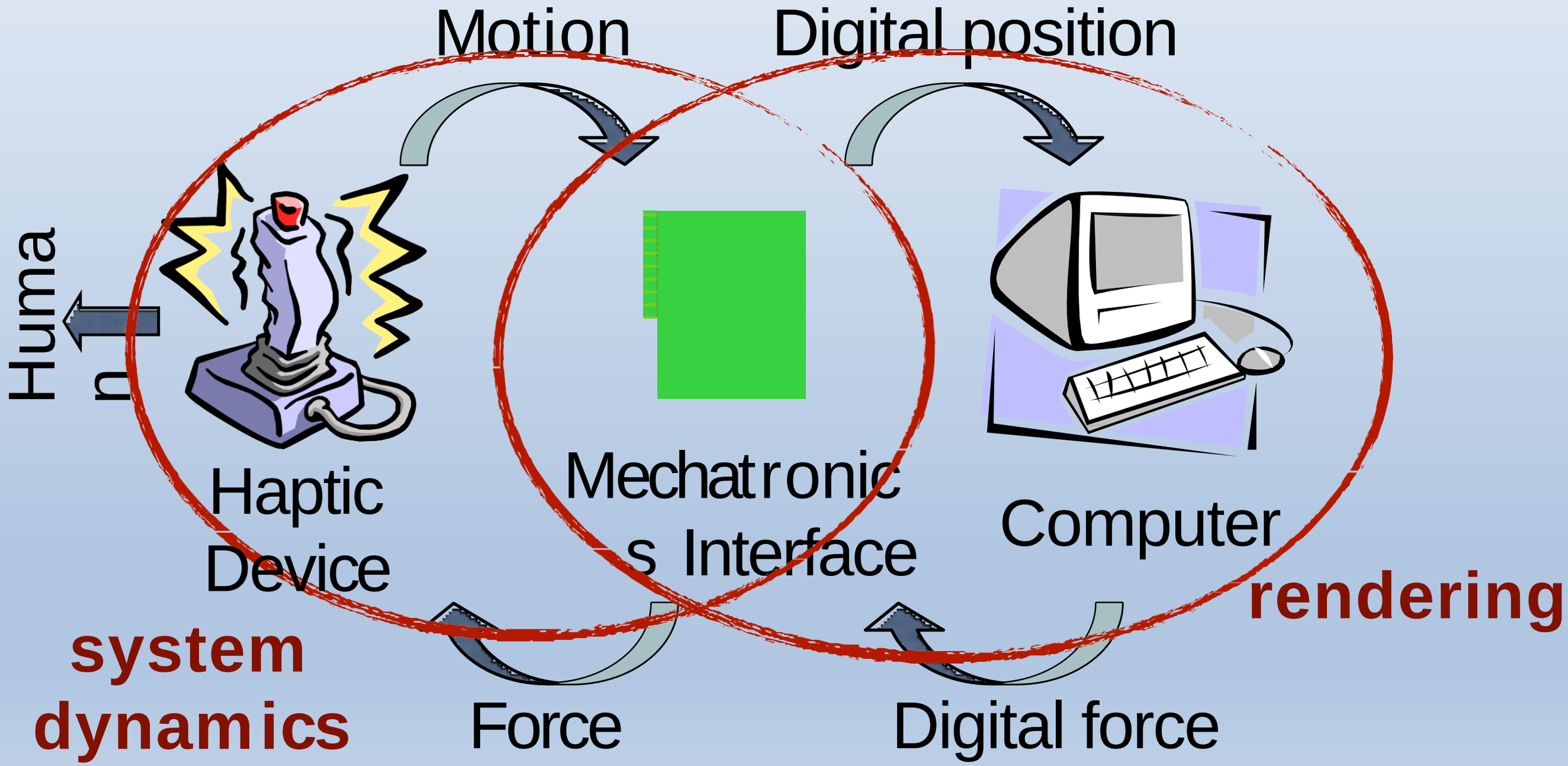
mechatronics basics

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a kinesthetic haptic system



a kinesthetic haptic system



motion signals

handle (m) → drum or motor (rad) → position sensor → computer or board → fwd kin. → virtual handle (m)

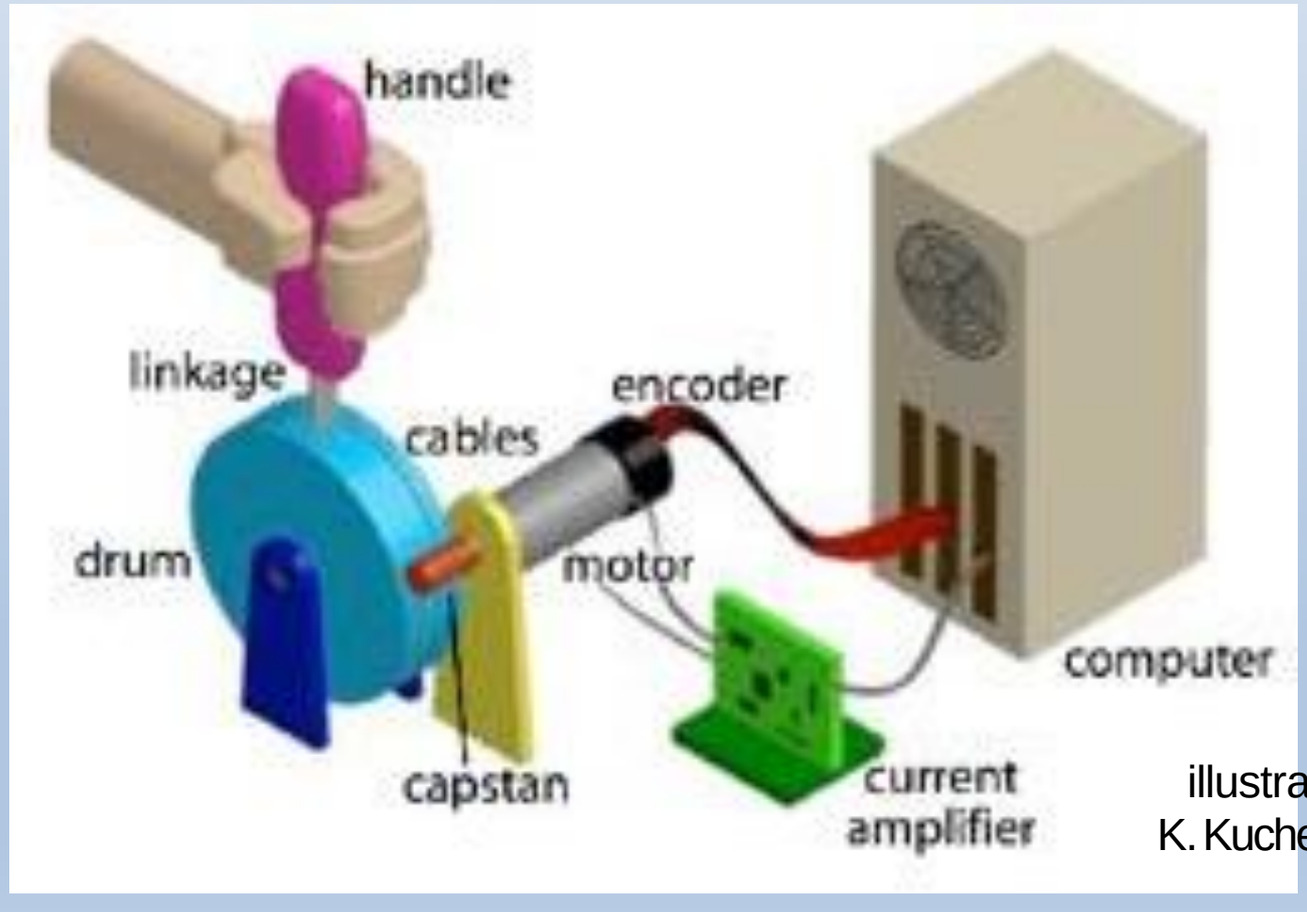
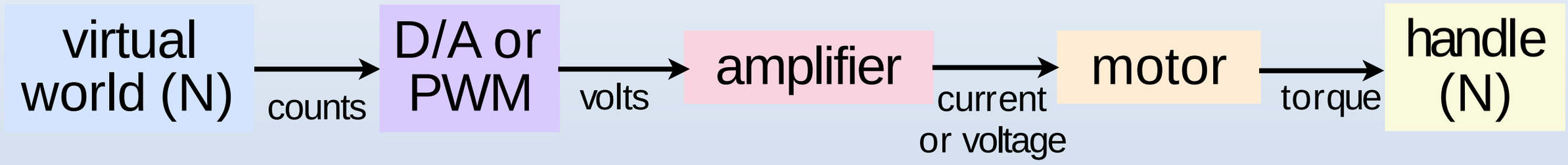


illustration by K. Kuchenbecker

force generation signals



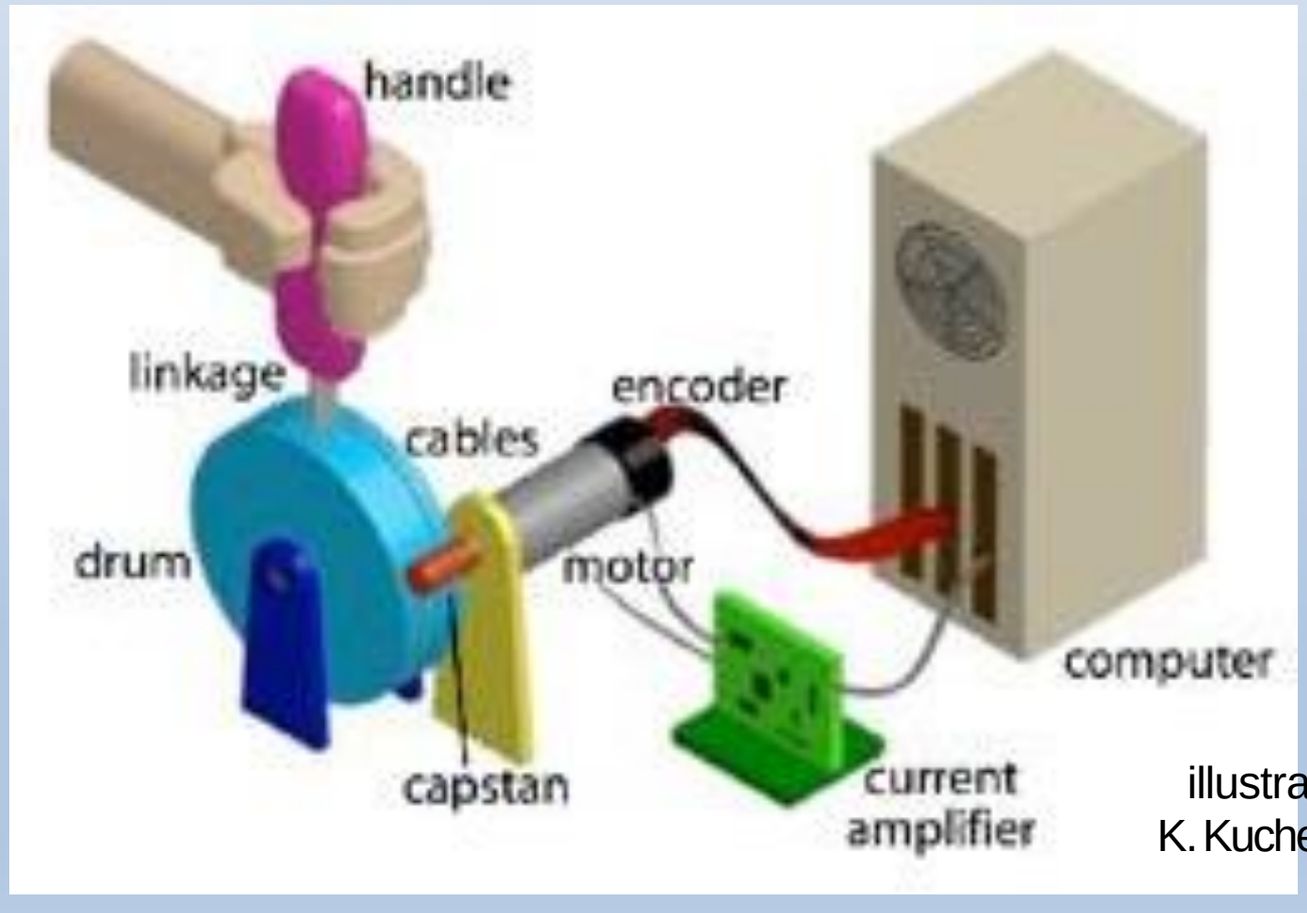


illustration by K. Kuchenbecker

Hapkit





